

SEMINAR 1: Mean-Field Optimal Control

(Massima Fornasier, Francesco Solombrino)

Zhang, Xiongtao

Seoul National University

zhangxiongtao1987@gmail.com

September 9, 2016

Table of Contents

1 Introduction

- Dynamic System
- Control Problem

2 Finite Agents

- Caratheodory Solution Existence
- Existence of Optimal Control

Cucker-Smale Model

Felipe Cucker and Steve Smale (2007)

CS model

$$\begin{cases} \frac{d}{dt}x^i = v^i, & t > 0, \quad 1 \leq i \leq N, \\ \frac{d}{dt}v^i = \frac{K}{N} \sum_{j=1}^N \hat{\psi}(|x^j - x^i|) (v^j - v^i), \end{cases} \quad (1)$$

- x_i : position of i -th particle,
- v_i : velocity of i -th particle,
- $\hat{\psi}$: communication function,
- K : coupling strength,
- N : number of particles.

Kinetic CS Model

For large numbers of particles, say $N \rightarrow +\infty$, PDE can describe the system.

Kinetic CS Model (formal by BBGKY)

$$\begin{cases} \partial_t f + v \partial_x f + \partial_v (\kappa_1 L(f) f) = 0, \\ L(f) = \int_{\mathbb{R}^{2d}} \psi(|x - y|) (v^* - v) f(v^*, y, t) dv^* dy. \end{cases} \quad (2)$$

$f(x, v, t)$ denotes the density of particles at position x , with velocity v , at time t .

Mean Field Limit

How to describe the limit process? More precisely, how to describe the connection

$$(x_i(t), v_i(t)) \xrightarrow{N \rightarrow +\infty} f(x, v, t).$$

- Put $(x_i(t), v_i(t))$ and $f(x, v, t)$ in the **same framework (space)**.
- Define a proper **metric d** on the space so that the distance between $(x_i(t), v_i(t))$ and $f(x, v, t)$ can be measured.
- Prove the **distance between $(x_i(t), v_i(t))$ and $f(x, v, t)$ tends to 0**, when N tends to ∞ .

Measure Valued Solution (Space)

Definition

(S. Ha, J. Liu 2009) For $T \in [0, T)$, let $\mu \in L^\infty([0, T); \mathcal{M}(\mathbb{R}^{2d}))$ be a measure valued solution with initial $\mu_0 \in \mathcal{M}(\mathbb{R}^{2d})$ iff μ satisfies

① (μ_t, g) is continuous as a function of t , for $g \in C_0(\mathbb{R}^{2d})$.

② μ satisfies the integral equation: $g \in C_0^1(\mathbb{R}^{2d} \times [0, T))$

$$(\mu_t, g(\cdot, \cdot, t)) - (\mu_0, g(\cdot, \cdot, 0)) = \int_0^t (\mu_s, \partial_s g + v \cdot \nabla_x g + F \cdot \nabla_v g) ds,$$

where $F(x, v, \mu_s) := - \int_{\mathbb{R}^{2d}} \psi(|x - y|)(v - v_*) \mu_s(dy, dv_*)$.

(S. Ha, J. Liu 2009)

- (ODE) The solution of CS model (x_i, v_i) can be viewed as a measure valued solution of the kinetic CS model as

$$\mu_t^N := \frac{1}{N} \sum_i \delta(x - x_i(t)) \delta(v - v_i(t)).$$

- (PDE) The classical solution of kinetic CS model is a measure valued solution.

Wasserstein Distance (Metric)

Define $\mathcal{P}_1$ to be of measures μ such that for some $x_0 \in X$,

$$\int_X |x - x_0| d\mu(x) < \infty.$$

Then, the Wasserstein-1 distance between the two probability measures μ and ν in $\mathcal{P}_1(X)$ is defined by

Definition

$$W_1(\mu, \nu) := \inf_{\gamma \in \Gamma(\mu, \nu)} \int_{X \times X} |x - y| d\gamma(x, y), \quad (3)$$

where $\Gamma(\mu, \nu)$ denotes the collection of all measures in $X \times X$ with the marginals μ and ν .

Rigorous Mean Filed Limit

The rigorous proof follows the results below:

S. Ha, J. Liu 2009

- Suppose $\mu, \nu \in L^\infty([0, T]; \mathcal{M}(\mathbb{R}^{2d}))$ are two measure value solution of kinetic CS model with initial datum (μ_0, ν_0) . (μ_0, ν_0) are compact supported with finite moments up to 2. Then there exists a constant $C(T)$ such that

$$W_1(\mu_t, \nu_t) \leq C(T)W_1(\mu_0, \nu_0).$$

- For any $\mu_0 \in \mathcal{M}(\mathbb{R}^{2d})$, there exists a sequence μ_0^N such that

$$\lim_{N \rightarrow \infty} W_1(\mu_0, \mu_0^N) = 0.$$

Table of Contents

1 Introduction

- Dynamic System
- Control Problem

2 Finite Agents

- Caratheodory Solution Existence
- Existence of Optimal Control

ODE Model

$$\begin{cases} \frac{d}{dt}x^i = v^i, & t > 0, \quad 1 \leq i \leq N, \\ \frac{d}{dt}v^i = (H * \mu_N)(x_i, v_i) + f(t, x_i, v_i), & i = 1, \dots, N, \quad t \in [0, T], \\ \mu_N = \frac{1}{N} \sum_{j=1}^N \delta_{(x_j, v_j)}. \end{cases} \quad (4)$$

μ_N is the empirical measure supported on the agents $(x_i, v_i) \in \mathbb{R}^{2d}$,
 $f(t, x_i, v_i)$ is the a controller and is vector valued function i.e

$$f(t, x, v) : \mathbb{R} \times \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}^d.$$

Cost Function

For any controller $f(t, x, v)$ of the system, we can define the cost functional:

Cost Function

$$\mathcal{E}_\psi^N(f) := \int_0^T \int_{\mathbb{R}^{2d}} (L(x, v, \mu_N(t)) + \psi(f(t, x, v))) d\mu_N(t)(x, v) dt,$$

where we require

- $\psi : \mathbb{R}^d \rightarrow [0, +\infty]$ is a nonnegative convex function.
- $H : \mathbb{R}^{2d} \rightarrow \mathbb{R}^d$ is a sublinear and locally Lipschitz continuous interaction kernel.
- $L : \mathbb{R}^{2d} \times \mathcal{P}_1(\mathbb{R}^{2d}) \rightarrow \mathbb{R}_+$ is a continuous function with respect to the product topology.

Cucker-Smale (example)

- $\psi(\cdot) = \gamma |\cdot|$.
- $H(x, v) = a(|x|)v$, where a is a non-increasing smooth positive function.
- $L(x, v, \mu_N) = |v - (\int_{\mathbb{R}^{2d}} w d\mu_N(y, w))|^2 = |v - \frac{1}{N} \sum_{j=1}^N v_j|^2$.

CS Model with Controller

$$\begin{cases} \frac{d}{dt} x^i = v^i, & t > 0, \quad 1 \leq i \leq N, \\ \frac{d}{dt} v^i = \frac{K}{N} \sum_{j=1}^N a(|x^j - x^i|) (v^j - v^i) + f(t, x_i, v_i). \end{cases} \quad (5)$$

Cost Function

$$\mathcal{E}_{\psi}^N(f) = \int_0^T \frac{1}{N} \sum_{i=1}^N \left(|v - \frac{1}{N} \sum_{j=1}^N v_j|^2 + \gamma |f(t, x_i, v_i)| \right) dt.$$

Optimal Control Problem

Let $F([0, T])$ be the collection of all controller $f(t, x, v)$, if there exists a control f_∞ such that $(x_i^\infty(t), v_i^\infty(t))$ solves the equation

Solution Exists

$$\begin{cases} \frac{d}{dt}x^i = v^i, & t > 0, \quad 1 \leq i \leq N, \\ \frac{d}{dt}v^i = (H * \mu_N)(x_i, v_i) + f_\infty(t, x_i, v_i), & i = 1, \dots, N, \quad t \in [0, T]. \end{cases} \quad (6)$$

Minimum Cost

$$\mathcal{E}_\psi^N(f_\infty) = \min_{f \in F([0, T])} \mathcal{E}_\psi^N(f).$$

When N tends to infinity, like the non-control problem, we have the PDE model

PDE

$$\mu_t + v \cdot \nabla_x \mu = \nabla_v \cdot [(H * \mu + f)\mu].$$

and the cost function defined as

Cost Function

$$\mathcal{E}_\psi(f) := \int_0^T \int_{\mathbb{R}^{2d}} (L(x, v, \mu(t)) + \psi(f(t, x, v))) d\mu(t)(x, v) dt,$$

Optimal Control problem

Let $F([0, T])$ be the collection of all controller $f(t, x, v)$, if there exists a control f_∞ such that μ_∞ is a measure valued solution of the equation

PDE

$$\mu_t + v \cdot \nabla_x \mu = \nabla_v \cdot [(H * \mu + f)\mu].$$

Moreover

Minimum Cost

$$\mathcal{E}_\psi(f_\infty) = \min_{f \in F([0, T])} \mathcal{E}_\psi(f).$$

To describe the mean-field limit we need to clarify

- The **existence** of the control problem in both ODE and PDE level.
- The existence of the **optimal control** f_∞ for both ODE and PDE model.
- Connect ODE and PDE solutions in the sense of measure valued solution and show the **mean-field limit**.

Table of Contents

1 Introduction

- Dynamic System
- Control Problem

2 Finite Agents

- Caratheodory Solution Existence
- Existence of Optimal Control

Definition

For a horizon time $T > 0$, and an exponent $1 \leq q < +\infty$, we fix a control bound function $\ell \in L^q(0, T)$. The class of admissible control functions $\mathcal{F}_\ell([0, T])$ is so defined: $f \in \mathcal{F}_\ell([0, T])$ if and only if

- $f : [0, T] \times \mathbb{R}^n \rightarrow \mathbb{R}^d$ is a **Caratheodory** function,
- $f(t, \cdot) \in W_{loc}^{1,\infty}(\mathbb{R}^{2d}, \mathbb{R}^d)$ for almost every $t \in [0, T]$, and
- $|f(t, 0)| + \text{Lip}(f(t, \cdot), \mathbb{R}^d) \leq \ell(t)$ for almost every $t \in [0, T]$.

Where Caratheodory function means

Caratheodory Function

- For fixed x , $f(\cdot, x)$ is **measurable** in Ω_x , the cross section of x .
- For fixed t , $f(t, \cdot)$ is **continuous** in Ω_t , the cross section of t .

Caratheodory Differential Equation and Solution

Consider the differential equation

$$\dot{y}(t) = g(t, y(t)), \quad t \in [0, T], \quad y \in \Omega \subseteq \mathbb{R}^n.$$

Definition

We call the equation is Caratheodory if $g(t, y)$ is a Caratheodory function.

Definition

For $0 < \tau \leq T$, we call $y : [0, \tau] \rightarrow \Omega$ a Caratheodory solution to the equation on $[0, \tau]$ if

- y is absolutely continuous,
- y satisfies the equation a.e in $[0, \tau]$.

Theorem

(Filippov A.1)

- Consider Caratheodory equation, if there exists a function $m \in L^1((0, T))$ such that

$$|g(t, y)| \leq m(t)$$

for a.e $t \in [0, T]$ and $y \in \Omega$. Then given $y_0 \in \Omega$, there exists $0 < \tau \leq T$ and a Caratheodory solution $y(t)$ on $[0, \tau]$ with initial data $y(0) = y_0$.

- More over, if there exists $\ell \in L^1((0, T))$ such that

$$|g(t, y_1) - g(t, y_2)| \leq \ell(t)|y_1 - y_2|.$$

Then the Caratheodory solution is unique on $[0, \tau]$.

Theorem

(A.2) Consider an interval $[0, T]$. Assume there exists a function $m \in L^1((0, T))$ such that

$$|g(t, y)| \leq m(t)(1 + |y|)$$

for a.e. $t \in [0, T]$ and $y \in \mathbb{R}^n$. Then for any $y_0 \in \mathbb{R}^n$, there exists a solution $y(t)$ defined on $[0, T]$ with $y(0) = y_0$. The solution satisfies

$$|y(t)| \leq \left(|y_0| + \int_0^t m(s) ds \right) e^{\int_0^t m(s) ds}.$$

Proof:

Consider our equation

$$\begin{cases} \frac{d}{dt}x^i = v^i, & t > 0, \quad 1 \leq i \leq N, \\ \frac{d}{dt}v^i = (H * \mu_N)(x_i, v_i) + f(t, x_i, v_i), & i = 1, \dots, N, \quad t \in [0, T], \\ \mu_N = \frac{1}{N} \sum_{j=1}^N \delta_{(x_j, v_j)}. \end{cases} \quad (7)$$

It is easy to check the right hand side satisfies previous theorem.

Proof:

Table of Contents

1 Introduction

- Dynamic System
- Control Problem

2 Finite Agents

- Caratheodory Solution Existence
- Existence of Optimal Control

Outline

- Find a control f_∞ .
- Prove the corresponding control problem admits a Caratheodory solution.
- Prove f_∞ is a minimizer

$$\mathcal{E}_\psi^N(f_\infty) = \min_{f \in F([0, T])} \mathcal{E}_\psi^N(f).$$

Idea of Construction of f_∞

As the cost functional is non-negative, for any control $f(t, x, v)$, there is a natural lower bound zero for $\mathcal{E}_\psi^N(f)$. Thus there exists a minimizing sequence $\mathcal{E}_\psi^N(f_k)$ such that

$$\lim_{k \rightarrow \infty} \mathcal{E}_\psi^N(f_k) = \inf \mathcal{E}_\psi^N.$$

In order to apply compactness to obtain the optimal control, we need the structure of the space of controllers i.e $\mathcal{F}_\ell([0, T])$.

Description of $\mathcal{F}_\ell([0, T])$

Remark 2.2

Suppose $f \in \mathcal{F}_\ell([0, T])$, then we have $f(t)$ is measurable for any fixed t and

$$f \in L^q((0, T), W^{1,p}(\Omega, \mathbb{R}^d))$$

for every open bounded subset $\Omega \subset \mathbb{R}^n$, $1 < p < +\infty$ and q is the integrability exponent of ℓ .

Proof:

Description of $\mathcal{F}_\ell([0, T])$

Remark 2.3

Consider the space

$$C_{\ell, \Omega} := \{f \in L^q((0, T), W^{1,p}(\Omega, \mathbb{R}^d)) : |f(t, 0)| + \text{Lip}(f(t, \cdot), \mathbb{R}^d) \leq \ell(t)\},$$

a.e. $t \in [0, T]$. If $f \in C_{\ell, \Omega}$ for all $\Omega \subset \mathbb{R}^n$, then $f \in \mathcal{F}_\ell$.

We only need to show f is Caratheodory. $f(t, \cdot)$ continuous because of the property of $C_{\ell, \Omega}$. Apply the equivalence of weak measurable and measurable as in Remark 2.2 to have $\langle f(t), x_0 \rangle = f(x_0, t)$ is measurable.

Theorem 2.5

Let X be a reflexive and separable Banach space. Let (f_j) be a sequence of functions in $L^q((0, T), X)$. Suppose there exists $m \in L^q([0, T])$ such that $\|f_j(t)\|_X \leq m(t)$ for a.e $t \in [0, T]$. Then there exists a subsequence (f_{j_k}) and a function $f \in L^q([0, T])$ such that

$$\lim_{k \rightarrow \infty} \int_0^T \langle \phi(t), f_{j_k}(t, \cdot) - f(t, \cdot) \rangle = 0,$$

for all $\phi \in L^{q'}((0, T), X')$ and

$$w - \lim_{k \rightarrow \infty} \int_{t_1}^{t_2} f_{j_k}(t) dt = \int_{t_1}^{t_2} f(t) dt, \quad t_1, t_2 \in [0, T].$$

The integrals are in the sense of Bochner.

Proof:

Theorem 2.6

Let Ω be a bounded smooth and open subset of $\mathbb{R}^n$. Assume (f_j) be a sequence of functions in $L^q((0, T), W^{1,p}(\Omega, \mathbb{R}^d))$ such that

$$|f_j(t, 0)| + \text{Lip}(f_j(t, \cdot), \Omega) \leq \ell(t) \quad \text{a.d. } t \in [0, T].$$

Then there exists a subsequence (f_{j_k}) and a function $f \in L^q((0, T), W^{1,p}(\Omega, \mathbb{R}^d))$ such that

$$w - \lim_{k \rightarrow \infty} f_{j_k} = f.$$

weakly in $L^q((0, T), W^{1,p}(\Omega, \mathbb{R}^d))$ and

$$|f(t, 0)| + \text{Lip}(f(t, \cdot), \Omega) \leq \ell(t) \quad \text{a.d. } t \in [0, T].$$

Corollary 2.7

Assume (f_j) be a sequence of functions in $\mathcal{F}_\ell$ for a given function ℓ . Then there exists a subsequence and function $f \in \mathcal{F}_\ell$ such that

$$\lim_{k \rightarrow \infty} \int_0^T \langle \phi(t), f_{j_k}(t, \cdot) - f(t, \cdot) \rangle dt = 0,$$

for all $\phi \in L^{q'}((0, T), H^{-1,p}(\mathbb{R}^n, \mathbb{R}^d))$ such that $\text{supp}(\phi) \subset \Omega$ for all $t \in [0, T]$, where Ω is relative compact.

Now consider the minimizing sequence, then the weak limit is our construction of f_∞ .

Find (x_∞, v_∞)

Lemma 3.2

Let $f \in \mathcal{F}_\ell$ and (x, v) be the solution of control problem with initial data $(x(0), v(0))$. Then

$$v(t) \leq \{v(0) + (1 + \mathcal{X}(0)) \left(2CT + \int_0^t \ell(s) ds \right)\} e^{(1+T) \int_0^t (2C + \ell(s)) ds}.$$

Where

$$v(t) = \max_N |v_i|, \quad \mathcal{X}(t) = \max_N |x_i|.$$

Then the trajectory $(x(t), v(t))$ is uniformly bounded.

Find (x_∞, v_∞)

With the equi-boundedness and equi-absolute continuity of $v_i^k(t)$, we can imply equi-Lipschitz of $x_i^k(t)$ and equi-integrability of $\dot{v}_i^k(t)$. Then by [Ascoli-Arzelà](#) theorem there exists a subsequence and an absolutely continuous trajectory $(x(t), v(t))$ such that

- $x_i^k \rightarrow x_i, \quad t \in [0, T],$
- $v_i^k \rightarrow v_i, \quad t \in [0, T],$
- $\dot{x}_i^k \rightarrow \dot{x}_i, \quad t \in [0, T],$
- $\dot{v}_i^k \rightharpoonup \dot{v}_i, \quad L^q([0, T]).$

Caratheodory Solution

Now we are going to show the constructed (x_i, v_i) is a Caratheodory solution. As we already show (x_i, v_i) are absolute continuous, we only need to show that

ODE

$$\begin{cases} \frac{d}{dt}x^i = v^i, & t > 0, \quad 1 \leq i \leq N, \\ \frac{d}{dt}v^i = (H * \mu_N)(x_i, v_i) + f_\infty(t, x_i, v_i), & i = 1, \dots, N, \quad t \in [0, T], \\ \mu_N = \frac{1}{N} \sum_{j=1}^N \delta_{(x_j, v_j)}. \end{cases} \quad (8)$$

We have four terms to deal with

Convergence for Different Terms

- For the first equation, we apply uniformly convergence and can obtain the results.
- Lemma A.7 shows $H * \mu_N^k(x_i^k, v_i^k) \rightarrow H * \mu_N(x_i, v_i)$.
- $\dot{v}_i^k \rightharpoonup \dot{v}_i, \quad L^q([0, T])$.
- The control term $f(t, x, v)$ is the difficult term.

Caratheodory Solution

For the second equation it is equivalent to consider

$$\xi \cdot \int_0^{\bar{t}} \dot{v}_i(t) dt = \xi \cdot \int_0^{\bar{t}} [(H * \mu_N) + f(t, x_i(t), v_i(t))] dt.$$

From previous analysis, we only need to prove

$$\lim \xi \cdot \int_0^{\bar{t}} f_k(t, x_i^k(t), v_i^k(t)) dt = \xi \cdot \int_0^{\bar{t}} f(t, x_i(t), v_i(t)) dt.$$

Proof:

Thank You!